

# SHETH MVLU COLLEGE

## SUBJECT:- WSN

**AIM:Simulation of TDMA-based MAC Protocol**

**Simulate TDMA slot allocation and compare energy efficiency with CSMA.**

**CODE:-**

```
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 4
set val(rp) DSDV
set val(x) 500
set val(y) 500
```

```
set ns [new Simulator]
```

```
set tracefd [open tdma.tr w]
$ns trace-all $tracefd
```

```
set namfd [open tdma.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)
```

```
set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
```

```
create-god $val(nn)
```

```
set chan_1_ [new $val(chan)]
```

```
$ns node-config \
  -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen $val(ifqlen) \
  -antType $val(ant) \
  -propType $val(prop) \
  -phyType $val(netif) \
  -channel $chan_1_ \
  -topoInstance $topo \
  -agentTrace ON \
  -routerTrace ON \
  -macTrace ON
```

```
for {set i 0} {$i < $val(nn)} {incr i} {
  set node_($i)[ns node]
}
```

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```
$node_(0) set X_ 50  
$node_(0) set Y_ 250  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 150  
$node_(1) set Y_ 250  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 350  
$node_(3) set Y_ 250  
$node_(3) set Z_ 0
```

```
for {set i 0} {$i < $val(nn)} {incr i} {  
    $ns initial_node_pos node_$i 20  
}
```

```
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

```
set null0 [new Agent/Null]  
$ns attach-agent $node_(3) $null0
```

```
$ns connect $udp0 $null0
```

```
set cbr0 [new Application/Traffic/CBR]  
$cbr0 set packetSize_ 512  
$cbr0 set interval_ 0.05  
$cbr0 attach-agent $udp0
```

```
set udp1 [new Agent/UDP]  
$ns attach-agent $node_(1) $udp1
```

```
set null1 [new Agent/Null]  
$ns attach-agent $node_(3) $null1
```

```
$ns connect $udp1 $null1
```

```
set cbr1 [new Application/Traffic/CBR]  
$cbr1 set packetSize_ 512  
$cbr1 set interval_ 0.05  
$cbr1 attach-agent $udp1
```

```
set udp2 [new Agent/UDP]  
$ns attach-agent $node_(2) $udp2
```

```
set null2 [new Agent/Null]
```

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```
$ns attach-agent $node_(3) $null2
```

```
$ns connect $udp2 $null2
```

```
set cbr2 [new Application/Traffic/CBR]
```

```
$cbr2 set packetSize_ 512
```

```
$cbr2 set interval_ 0.05
```

```
$cbr2 attach-agent $udp2
```

```
nsat1.0"cbr0 start"
```

```
nsat1.5"cbr0 stop"
```

```
nsat2.0"cbr1 start"
```

```
nsat2.5"cbr1 stop"
```

```
nsat3.0"cbr2 start"
```

```
nsat3.5"cbr2 stop"
```

```
nsat4.0"cbr0 start"
```

```
nsat4.5"cbr0 stop"
```

```
proc finish {} {
```

```
    global ns tracefd namfd
```

```
    $ns flush-trace
```

```
    close $tracefd
```

```
    close $namfd
```

```
    puts "Simulation Completed"
```

```
    exec nam tdma.nam &
```

```
    exit 0
```

```
}
```

```
$ns at 5.0 "finish"
```

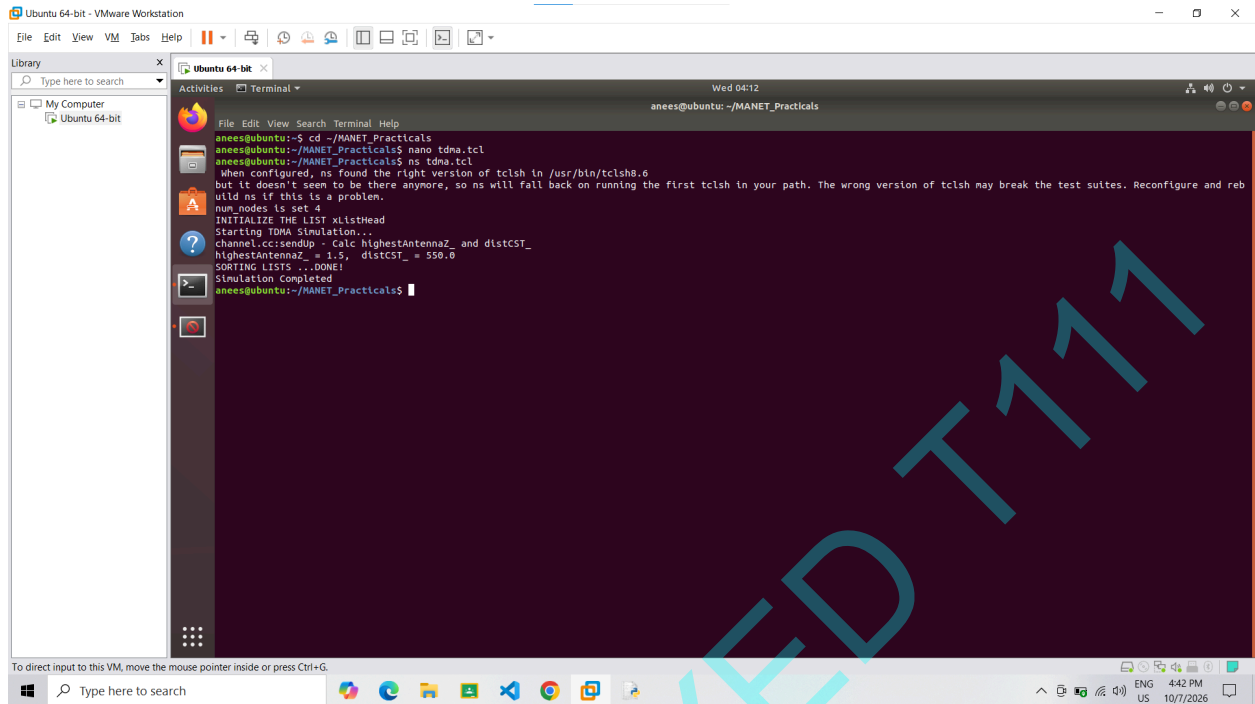
```
puts "Starting TDMA Simulation..."
```

```
$ns run
```

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OUTPUT:-



The screenshot shows a terminal window titled 'Ubuntu 64-bit' within a VMware Workstation environment. The user 'anees' is in the directory '~/MANET\_Practicals'. They have executed the following commands and received the following output:

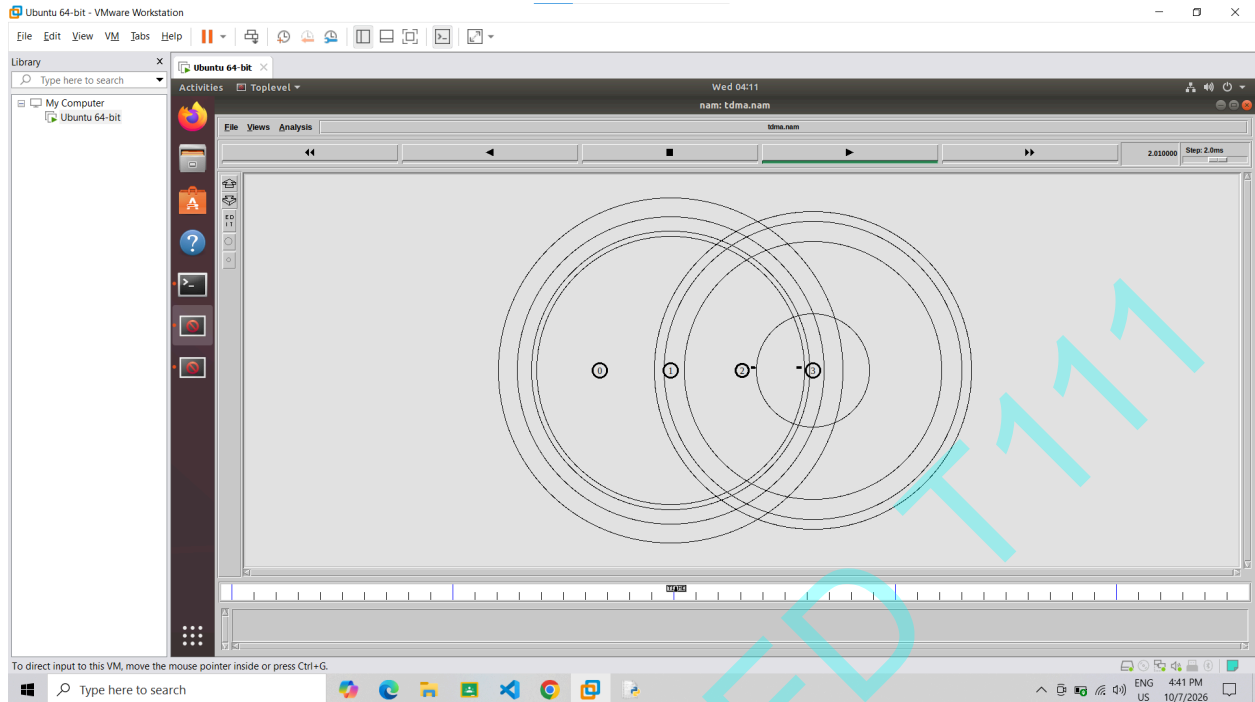
```
anees@ubuntu:~$ cd ~/MANET_Practicals
anees@ubuntu:~/MANET_Practicals$ nano tdma.tcl
anees@ubuntu:~/MANET_Practicals$ ns tdma.tcl
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tclsh in your path. The wrong version of tclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
INITIALIZE THE LIST xlistHead
Starting TDMA Simulation...
channel.cc:sendUp - Calc highestAntennaZ_ and distcST_
highestAntennaZ_ = 1.5, distcST_ = 550.0
SORTING LISTS ...DONE!
Simulation Completed
anees@ubuntu:~/MANET_Practicals$
```

A large, diagonal watermark reading 'ANEES SAYED T111' is overlaid across the entire image.

ANEES SAYED T111

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### Observation

- Four wireless nodes were created.
- Node 3 acted as the destination node.
- Nodes 0, 1, and 2 transmitted data in separate time slots.
- No simultaneous transmissions occurred.
- Packet collisions were avoided due to slot allocation.
- Data packets successfully reached the destination node.
- NAM animation clearly showed sequential transmissions.

### Result

The TDMA-style MAC protocol simulation was successfully implemented. Scheduled transmissions occurred according to assigned time slots, and packets were delivered successfully to the destination node.